1 Tesla Road, Austin TX 78725
https://tesla.com

November 21, 2025

Ms. Debbie-Anne A. Reese
Secretary
Federal Energy Regulatory Commission
888 First Street NE
Washington, D.C. 20426

PUBLIC COMMENTS

Submitted electronically via https://www.ferc.gov

Re: Comments from Tesla on the Proposed Advanced Notice of Proposed Rulemaking (Docket No. RM26-4-000)

Tesla, Inc. ("Tesla") appreciates the opportunity to respond to the Federal Energy Regulatory Commission's ("the Commission") Notice Inviting Comments on the proposed advance notice of proposed rulemaking ("proposed ANOPR") concerning the interconnection of large loads to the interstate transmission system.[1] We welcome a robust and thorough process to gather information from stakeholders to ensure any final rule, and its implementation, considers all affected parties.

Tesla recognizes the significant value of the principles of reform and offers the following comments and case study to inform the Commission's rulemaking process. These comments also highlight additional considerations where large loads and energy demand are concerned, including the role of distributed energy resources (DERs) and virtual power plants (VPPs).

Principles for Reform

While Tesla does not comment on each principle identified in the ANOPR, we agree with the overall intent of the Commission's proposed framework and offer the following considerations:

The fourth principle suggests load and hybrid facilities should be subject to study deposits, readiness requirements, and withdrawal penalties. Tesla is generally supportive of concepts that limit duplicative load interconnection requests; however, we believe that requirements should be instituted for large loads presumed to be data centers or crypto mining centers. Given load

[1] Interconnection of Large Loads to the Interstate Transmission System, Notice Inviting Comments, Docket No. RM26-4-000 (October 27, 2025).

requests have not been subjected to such mechanisms, a system should be established to provide credit back for study deposits, but perhaps through transmission credits.

Tesla strongly endorses the fifth principle, which notes that hybrid facilities should be studied based on the amount of injection and/or withdrawal rights requested to incentivize colocation and efficiency buildout. We interpret this reform as similar to the eleventh principle, involving utility serving large loads and hybrid facilities being responsible for transmission service based on withdrawal rights. We suggest that any additional load or battery energy storage systems (BESS) installed behind-the-meter (BTM) at large load facilities count towards the point of interconnection (POI) limit. Facilities should be responsible for ensuring that total demand remains below the contracted capacity, consistent with withdrawal right procedures. These principles are relevant to the Tesla Gigafactory Case study highlighted below.

The tenth principle raises that an existing generating facility seeking partial suspension to serve new load at the same location must undergo a system support resource (SSR)/reliability must run (RMR) type of study. Tesla believes that capacity additions behind the same POI that offset generation "loss" to serve load should be permitted with expedited reviews, assuming the system maintains consistent or similar POI import/export limitations. For example, when installing a data center co-located with an existing generator, we see potential to install fast-to-deploy energy storage to offset the system capacity loss or effective load carrying capability (ELCC).

The fourteenth principle suggests utilities serving large loads should meet all applicable NERC reliability standards and Open Access Transmission Tariff (OATT) provisions. It emphasizes the need to update procedures, agreements, and standards to ensure the reliability of the Bulk Electric System (BES). Tesla recommends establishing minimum interconnection standards for large loads, including ramp rate, load stability, and voltage ride-through rate as standardized with the process. These should be applied at the POI and with a technology-neutral lens. Similar performance expectations are being applied or developed in regions (e.g. SPP and ERCOT) and could serve as a foundation for national best practices. Additionally, if a load meets the minimum interconnection standards, additional information on its makeup, which may change over time, should not be required—especially at early stages in the load's development when it may not be available.

Case Study: Tesla's Gigafactory Texas

Gigafactory Texas is Tesla's 2,500-acre corporate headquarters and large-scale manufacturing facility near Austin, Texas. It plays a central role in our mission to accelerate sustainable abundance by producing battery materials, components, and electric vehicles, including Tesla's Cybertruck, Model Y, and future Cybercab. Gigafactory Texas also embodies our principles by integrating energy projects, such as the installation of 133MW of grid-connected Tesla Megapacks, our utility-scale BESS product.

In 2024, Tesla pursued interconnection for the Gigafactory Texas Megapack project, declaring the use case as back-up generation to load. The interconnection process bucket was unclear for this application and, therefore, ERCOT developed a pathway. In ERCOT, BESS are studied as generator resources, grid services that provide stability to the grid (i.e. voltage and frequency), or ancillary services. These projects are also required to meet additional milestones, including quarterly stability assessment (QSA) approvals, full project registration, energization, synchronization, and commissioning.

While contending with timeline concerns, Tesla reevaluated regulatory requirements and discovered that a non-exporting BESS back-up generator did not require ERCOT registration. This finding was initially validated by ERCOT's presentation to the Texas House of Representatives Committee on State Affairs[2] and later reinforced by ERCOT's Large Load Working Group in its "FAQ: Stakeholder Questions on Large-Loads."[3] In the absence of this discovery, delays in establishing BESS back-up generation at Gigafactory Texas would pose serious financial risk during a utility outage at Tesla's largest manufacturing facility.

This experience highlights a broader need to establish greater clarity, efficiency, and predictability in the interconnection process as large load growth driven by manufacturing facilities and data centers continues at a rapid pace.

Additional Consideration: Interconnection, DERs, and VPPs

Tesla is a leader in distributed energy resources (DERs) and Virtual Power Plants (VPPs) with more than 1 million distributed batteries installed and 250,000 (representing 1.5GW) participating in grid services programs. Distributed energy resources can provide considerable capacity to support high load growth. VPPs, which are coordinated fleets of DERs like batteries and electric vehicles, can be deployed in weeks and months, not the years-long construction timelines of conventional generation.

[2] "ERCOT Update," Texas House of Representatives, Committee on State Affairs (March 5, 2025).
[3] "FAQ: Stakeholder Questions on Large-Loads," Large Load Working Group, Electric Reliability Council of Texas (August 15, 2025).

In any proposed rule by the Commission, Tesla urges the recognition of DERs and VPPs as meaningful grid service tools that can support faster interconnection of large loads to the transmission system. VPPs should be considered for their potential to alleviate transmission network constraints and capacity needs associated with the interconnection of large loads. Tesla further urges the commission to look at constraints that today block the deployment of VPPs to provide accredited capacity. In 2020, the Commission issued Order No. 2222 directing grid operators to allow DER aggregations, such as VPPs, to participate in wholesale electricity markets. However, restrictive interpretations of Order No. 2222 continue to limit the participation of these resources. Currently, no interpretation of the Order practically enables the full value of an inverter-based DER. FERC must address challenges in 2222 implementations including: 1) metering and telemetry barriers, 2) device size thresholds, 3) state-level barriers, and particularly 4) the lack of pathways for DERs receiving retail rate credits. Tesla believes that there is a pathway to unblock VPPs while addressing concerns about double-counting.

Ensuring that DERs and VPPs are unblocked to provide accredited capacity and then allowing for the consideration of VPPs' capacity while considering the interconnection of large loads will unlock a new and massive category of untapped capacity to solve the growing pressure on the transmission system.

Conclusion

Tesla's status as a leader in the battery storage industry uniquely positions us to provide valuable insight to FERC during this process, and we welcome any opportunities to further discuss this important matter. Thank you for the consideration of these comments.

About Tesla, Inc.

Tesla's mission is to accelerate the world's transition to sustainable energy.[4] Tesla is a U.S. company, headquartered in Austin, Texas, and publicly traded under the ticker symbol "TSLA".[5] To accomplish its mission, Tesla designs, develops, manufactures, and sells high-performance fully electric vehicles ("EVs") and energy generation and storage systems, installs, and maintains such systems, and sells solar electricity.[6] Consistent with this effort, Tesla was

---

[4] *See*, Tesla, About, *available at* https://www.tesla.com/about.

[5] *See*, U.S. Securities and Exchange Commission ("SEC"), https://www.sec.gov/edgar/browse/?CIK=0001318605. *See also* Tesla's most recent Form 10-Q filed with the SEC, *available at* https://www.sec.gov/ix?doc=/Archives/edgar/data/0001318605/000162828025045968/tsla-20250930.htm.

[6] *See*, Tesla, Impact Report 2024 (June 30, 2025) *available at* https://www.tesla.com/impact.

recently ranked as the world leader in the transition to vehicle electrification.[7] Tesla also develops and deploys autonomy and real-world artificial intelligence at scale, in vehicles and robots, through the development of hardware and software, including Full Self-Driving (Supervised), Robotaxi, and the Optimus humanoid robot.[8] Tesla currently produces and sells five fully electric, zero emissions light-duty vehicles ("ZEVs"): including the best-selling car in the world (EV or otherwise), the Model Y compact sport utility vehicle.[9] Along with the Model Y, Tesla manufactures the Model 3 mid-sized sedan, Model S sedan, the Model X SUV, and Cybertruck.

In the United States, Tesla conducts manufacturing operations across multiple facilities. Collectively, Tesla's U.S. facilities support over 70,000 employees and generate billions of dollars of U.S. investment and economic activity each year. Specifically, in its facilities in Fremont, California, Tesla conducts vehicle manufacturing and assembly operations, as well as manufacturing lithium-ion battery cells, battery packs, vehicle seats, stampings, and castings. At Megafactory 1 in Lathrop, California, Tesla produces Megapack, a utility-scale grid storage battery.[10] At Gigafactory Texas in Austin, Tesla produces the Model Y and Cybertruck as well as cathode active materials, lithium-ion battery cells, battery packs, drive units, vehicle seats, stampings, and castings. Gigafactory Texas will also be home to Tesla's next-generation vehicle assembly line and production of the fully autonomous Cybercab. In Sparks, Nevada, Tesla produces drive units, battery packs, and energy storage products (Powerwall) at Gigafactory Nevada, and the fully electric Semi, a commercial heavy-duty vehicle, at a new nearby facility under construction, which is planned for completion in early 2026. At Gigafactory New York in Buffalo, New York, Tesla produces its DC-fast charging equipment for the Tesla Supercharger network, solar energy products, and power electronics. Additionally, Tesla manufactures, builds, and services highly automated, high-volume manufacturing machinery at its facility in Brooklyn Park, Minnesota, and operates a tool and die facility in Grand Rapids, Michigan. Each Tesla factory uses state-of-the-art industrial robotics and manufacturing equipment to build an abundant future—they are the machines that build the machines.

[7] *See*, ICCT, The Global Automaker Rating 2022: Who Is Leading the Transition to Electric Vehicles? (May 31, 2023) *available at* https://theicct.org/publication/the-global-automaker-rating-2022-may23/,.
[8] *See*, Tesla, AI & Robotics, *available at* https://www.tesla.com/AI.
[9] *See*, Green Car Reports, The Bestselling Vehicle on the Planet is an EV (Jan. 25, 2024) a*vailable at* https://www.greencarreports.com/news/1142104_tesla-the-bestselling-vehicle-on-the-planet-is-an-ev.
[10] *See*, Tesla, Megapack, *available at* https://www.tesla.com/megapack.